

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Placebo-controlled, Global Proof-of-concept Study to Evaluate the Efficacy, Safety, and Tolerability of BMS-986446, an Anti-MTBR Tau Monoclonal Antibody, in Participants With Early Alzheimer's Disease **Short Title/Acronym:** TargetTau-1 **Protocol Number:** CN008-0003 **NCT Number:** NCT06268886 **Sponsor Name:** Bristol-Myers Squibb Company **Investigational Product:** BMS-986446 (Anti-MTBR Tau Monoclonal Antibody) **Phase of Development:** Phase II **Medical Monitor:** Head of Global Submission Management - Clinical Trials (Bristol-Myers Squibb)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate clinical decline in patients with early Alzheimer's Disease treated with BMS-986446 compared with placebo, evaluated based on the mean change from baseline in the Clinical Dementia Rating Scale Sum of Boxes (CDR-SB) score at 76 weeks. **Secondary Objectives:** To assess the safety, tolerability, and efficacy of BMS-986446 through additional cognitive and clinical assessments, and to evaluate tau deposition and pharmacokinetic/pharmacodynamic responses.

2.2 Background & Rationale Alzheimer's disease (AD) is characterized pathologically by extracellular amyloid-beta plaques and intracellular neurofibrillary tangles containing aggregated tau proteins. The microtubule-binding region (MTBR) within the tau protein is critical for the pathogenic self-association of tau and its neuronal uptake and transmission. BMS-986446 is a novel humanized monoclonal antibody that targets the MTBR to inhibit the cell-to-cell spread of neurotoxic tau species, addressing a critical unmet need to slow disease progression in patients with early AD.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized in a 4:3:3 ratio (Placebo vs. BMS-986446 1400 mg vs. BMS-986446 4200 mg)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 475 participants **Number of Sites:** Global trial (Multiple sites globally, including Belgium and others) **Estimated Study Start:**

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 50 to 80 years. 2. Mild cognitive impairment (MCI) due to Alzheimer's disease or mild AD dementia consistent with NIA-AA core clinical criteria. 3. Global Clinical Dementia Rating (CDR) score of 0.5 to 1.0 and a CDR-Memory Box score ≥ 0.5 at screening and Baseline. 4. Objective impairment in episodic memory (≥ 1 standard deviation below age-adjusted mean in WMS-IV LM II) and an MMSE score ≥ 20 to 28 (inclusive). 5. Evidence of AD pathology.

4.2 Exclusion Criteria 1. Any evidence of a condition that may affect cognition other than AD. 2. Contraindications to PET imaging or inability to tolerate magnetic resonance imaging (MRI). 3. Geriatric Depression Scale (GDS) score ≥ 8 at screening. 4. Any serious medical condition that could affect the participant's safety or interfere with study assessments.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: BMS-986446 1400 mg or 4200 mg (or matching placebo) administered every 4 weeks. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 76-week double-blind treatment period (followed by an optional 96-week open-label extension).

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care medications for co-morbidities that do not impact cognitive assessments. **Prohibited:** Other investigational anti-amyloid or anti-tau therapies and medications known to significantly alter cognitive function during the trial window.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Cognitive Assessments (MMSE, CDR, WMS-IV LM II), PET/MRI Scans
Baseline	Day 0	Randomization, First IV Infusion, Clinical Baseline Evaluation
Interim	Weeks 12, 24, 48	IV Infusions (Q4W), Safety Labs, Cognitive & Clinical Assessments
End of Study	Week 76	Final Double-Blind IV Dose, Primary Endpoint Assessment (CDR-SB)
Safety Follow-up	Week 87	Final Vital Signs, Exit Interview, Safety Follow-up (11 weeks post-last dose)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Clinical decline evaluated based on the mean change from baseline in the Clinical Dementia Rating Scale Sum of Boxes (CDR-SB) score at 76 weeks. **Measurement Method:** Standardized Clinical Dementia Rating interview and scoring tool.

7.2 Secondary & Exploratory Endpoints 1. Safety, tolerability, and incidence of adverse events (AEs). 2. Cognitive and clinical assessments at 12, 24, and 48 weeks. 3. Changes in tau deposition as measured by PET imaging and biofluid biomarkers.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during IV infusions and scheduled clinical visits, with particular focus on neuroimaging abnormalities (e.g., ARIA) via scheduled MRIs. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for all severe or life-threatening occurrences. **Data Monitoring Committee (DMC):** Independent External Data Monitoring Committee to oversee patient safety and unblinded data at specified intervals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy (CDR-SB), and Safety Set for adverse events. **Sample Size Justification:** $N = 475$ participants, randomized 4:3:3, powered to detect a statistically significant reduction in clinical decline over 76 weeks. **Handling of Missing Data:** Mixed-Effect Model Repeated Measure (MMRM) for longitudinal continuous cognitive outcomes.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits for source data verification and assessment standard validation. **Audit Trail:** Centralized grading for cognitive assessments and centralized reading for PET/MRI imaging to ensure consistency.

11. Study Identifiers & Governance

Primary ID: CN008-0003 **Registry Number:** NCT06268886 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** TargetTau-1 **Official Regulatory Title:** A Randomized, Double-blind, Placebo-

controlled, Global Proof-of-concept Study to Evaluate the Efficacy, Safety, and Tolerability of BMS-986446, an Anti-MTBR Tau Monoclonal Antibody, in Participants With Early Alzheimer's Disease

12. Clinical Framework (Alzheimer's Specific)

Primary Condition: Early Alzheimer's Disease (MCI and Mild AD Dementia) **Diagnosis Threshold:** MMSE score ≥ 20 , CDR score 0.5 to 1.0, Evidence of AD pathology **Medical Methodology:** Anti-MTBR tau monoclonal antibody infusion therapy **Optimization Target:** Slowing of cognitive and functional decline (Targeting Tau propagation)

13. Study Procedures & Methodology

Management Pathway: 4-weekly Intravenous (IV) infusions in a clinical setting **Education Protocol (HCP):** Site rater training and certification for cognitive tests (CDR-SB, WMS-IV LM II) **Education Protocol (Patient):** Caregiver involvement required for accurate CDR-SB reporting and safe transportation **Quality Audit Mechanism:** Centralized independent review of CDR-SB scores and neuroimaging scans (PET/MRI)

14. Observation & Follow-Up Schedule

Total Duration: 95 weeks (76 weeks of treatment + 11 weeks safety follow-up, plus screening) **Onsite Visit Cadence:** Every 4 weeks for IV infusion; comprehensive assessments at Weeks 12, 24, 48, 76 **Remote Monitoring Frequency:** As required for safety follow-up **Checklist Review Interval:** Cognitive/clinical assessments formally reviewed at Weeks 12, 24, 48, and 76

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				